MAIS PAC

IPhO 03.02: Uncertainty Analysis · Lecture + Problems

Uncertainty Analysis

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

7 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. Why uncertainties are not optional

A measurement without an uncertainty cannot be compared with a theory, with another measurement, or with a specification. "The resistance is 10.2 Ω" says nothing; "the resistance is (10.2 ± 0.1) Ω" says that a claim of 10.5 Ω is excluded and a claim of 10.25 Ω is not.

Two things must be distinguished at all times:

- The **uncertainty** of a single reading (resolution, reading error, calibration).
- The **spread** of a set of repeated readings, which tells you how precise the **procedure** is.

2. The statistics of repeats

If a quantity is measured N times with readings x_i, the best estimate is the mean

$$x = (1)/(N)\Sigma\ x_i, \tag{1}$$

and the spread is described by the standard deviation

$$s = \sqrt{((\Sigma\ (x_i - x)^2)/(N-1))}. \tag{2}$$

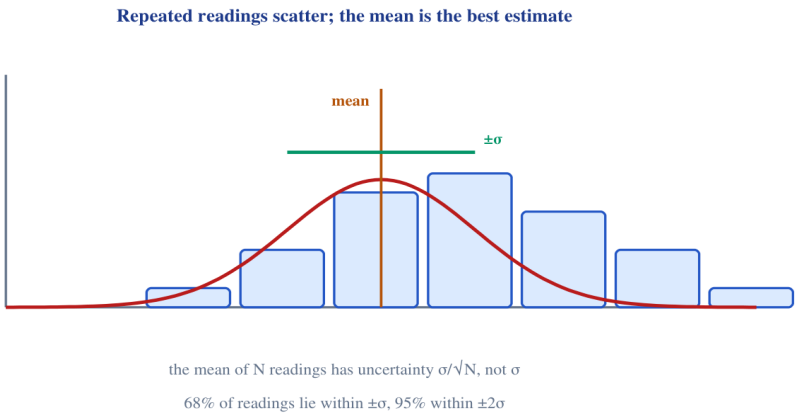


Figure 1. The random part of an experiment shows up as a spread. The width of that spread is a measurement in its own right: it tells you how precisely you measured, and it shrinks as 1/√N when you average.

The uncertainty of the **mean** is smaller than the spread of individual readings:

$$\sigma_x = (s)/(\sqrt{()}N). \tag{3}$$

That is the 1/√()N law: quadrupling the number of readings only halves the uncertainty. It is worth knowing because it tells you when to stop repeating and improve the apparatus instead.

For a Gaussian distribution, 68% of readings lie within ±σ of the mean, 95% within ±2σ, and 99.7% within ±3σ. In an exam, an uncertainty quoted as ±Δ is usually intended as a rough 1σ figure.

3. Propagation in one line

For f(x,y,) with independent uncertainties, the honest combined uncertainty is

$$\Delta\ f = \sqrt{(((f)/(x)\Delta\ x)^2 + ((f)/(y)\Delta\ y)^2)}, \tag{4}$$

which reduces to three memorisable rules:

- Sums and differences: **absolute** uncertainties add in quadrature.
- Products and quotients: **fractional** uncertainties add in quadrature.
- Powers: fractional uncertainty multiplied by the exponent, $\Delta\ f/f = |n|,\Delta\ x/x$.

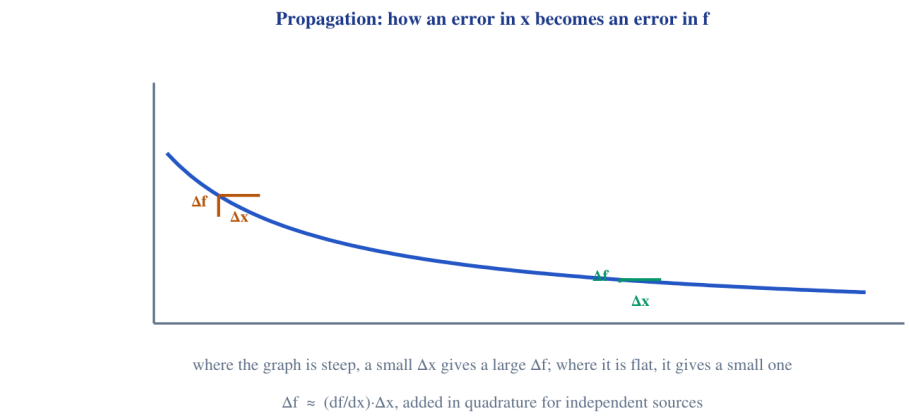


Figure 2. The derivative is the exchange rate between an error in the input and an error in the output. For a power law it turns the rules into one sentence: fractional errors add, multiplied by the exponents.

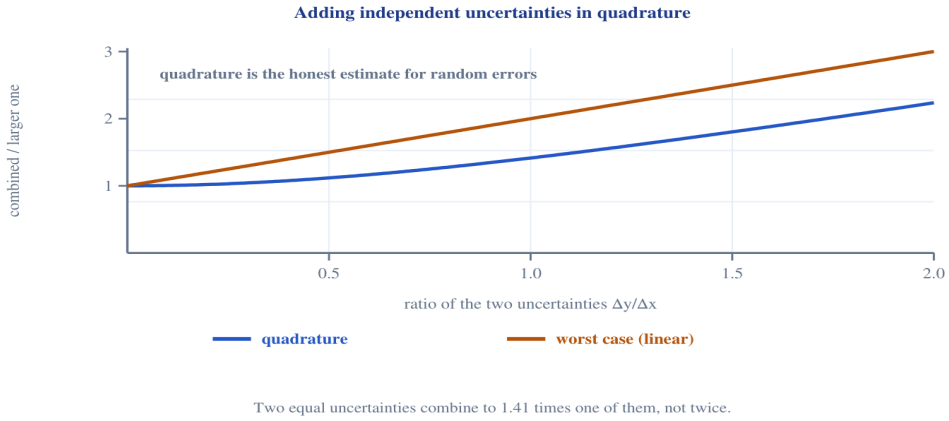


Figure. Two equal uncertainties combine to 1.41 times one of them, not twice.

Interactive experiment: Propagation through a power law — open the online handout to use it.

A worked example makes the pattern clear. For the density of a cylinder, $\rho = 4m/(\pi\, d^2\, h)$, the fractional uncertainty is

$$(\Delta\rho)/(\rho) = \sqrt{(((\Delta\ m)/(m))^2 + (2(\Delta\ d)/(d))^2 + ((\Delta\ h)/(h))^2)}, \tag{5}$$

so a 1% error in the diameter contributes 2% — the single most common reason a density experiment misses its target.

4. Absolute versus relative

Absolute and relative uncertainty

measurement	absolute	relative
L = 25.00 m	± 0.05 m	0.2 %
L = 2.500 m	± 0.005 m	0.2 %
L = 0.025 m	± 0.005 m	20 %
t = 120.0 s	± 0.2 s	0.17 %
t = 1.2 s	± 0.2 s	17 %

a small object measured with a good instrument can still have a poor relative uncertainty

quote absolute uncertainty for a single value, relative when combining measurements

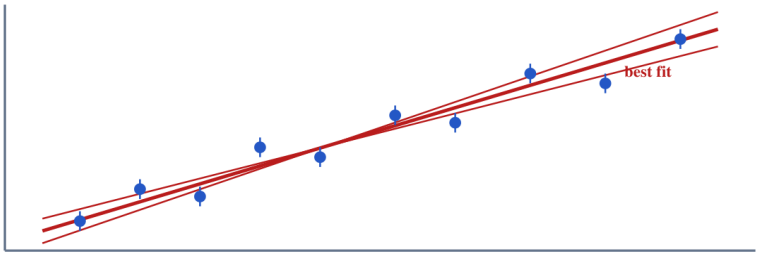
Figure 4. The same absolute uncertainty can be negligible or fatal depending on the size of the quantity. That is why the choice of instrument must be matched to the size of what you are measuring.

The same instrument gives the same absolute uncertainty whatever the size of the object, so the **relative** uncertainty grows as the object shrinks. That single fact decides which instrument to use: a micrometer on a 2 {mm} wire gives 0.5%, the same micrometer on a 2 {cm} rod gives 0.05%, and a ruler on the same wire would give 50%.

5. Graphs, gradients and intercepts

The most powerful analysis technique in the practical exam is linearisation: arrange the theory into the form y = mx + c, plot, and read the gradient and intercept.

A best-fit line through scattered points: the gradient is the measurement



gradient = Δy/Δx from the line, never from two scattered points

the dashed lines show the steepest and shallowest lines consistent with the error bars

Figure 3. Drawing the line is a way of averaging all the points with proper weights. The intercept often carries physics too: a non-zero intercept where theory demands zero is a systematic error.

- Draw the line of best fit by eye through the scatter, not by joining the first and last points.
- Estimate the **uncertainty in the gradient** by drawing the steepest and shallowest lines that are still consistent with the error bars; half the difference is the uncertainty.
- An intercept that theory says should be zero but is not is a **systematic error** in disguise — a zero error, a background, or an effect you forgot.

Examples worth learning: T^2 against L for a pendulum (gradient $4\pi^2/g$), 1/v against 1/u for a lens (intercept -1/f), ln I against x for absorption (gradient -μ), V against I for a cell (intercept the emf, gradient -r).

6. Reporting a result

The conventions that earn marks:

- Round the uncertainty to **one significant figure** (two if the first digit is 1).
- Match the value to the same decimal place: (9.81 ± 0.02) {m,s^(-2)}, not (9.8123 ± 0.02).
- Include units on both, and state what the uncertainty means (1σ, resolution, or range).
- For a discrepancy, quote it in units of the combined uncertainty: |x_1-x_2|/√(Δ x_1²+Δ x_2²). A ratio below about 2 is consistent; a ratio above 3 needs explaining.

The uncertainty checklist. Write down every measured quantity with its uncertainty before doing any algebra → convert to fractional uncertainties → combine with the rules (sums add absolute, products add fractional, powers scale fractional) → propagate into the final quantity, including any squared or cubed variables → round to one significant figure in the uncertainty and match the value → compare your result with the expected value in units of the uncertainty and comment if the difference is large.

7. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Mean and standard deviation. Five readings of a length in mm are 12.4, 12.6, 12.5, 12.3, 12.7. (a) Find the mean. (b) Find the standard deviation. (c) Find the uncertainty of the mean and quote the result correctly.

Worked solution →

[1] Problem 2 — Simple propagation. A rectangle has sides a = (4.0 ± 0.1) {cm} and b = (6.0 ± 0.2) {cm}. (a) Find the perimeter with its uncertainty. (b) Find the area with its uncertainty. (c) Explain why the area has a larger percentage uncertainty than either side.

Worked solution →

[2] Problem 3 — A density measurement. A metal cylinder has mass (84.2 ± 0.1) {g}, diameter (20.0 ± 0.1) {mm} and length (30.0 ± 0.2) {mm}. (a) Find the density. (b) Find the percentage uncertainty in the density. (c) State which measurement dominates and what improvement would help most.

Worked solution →

[2] Problem 4 — Timing and the pendulum. The time for 50 oscillations is (100.4 ± 0.2) {s} and the length is (1.000 ± 0.002) {m}. (a) Find the period with its uncertainty. (b) Find g with its uncertainty. (c) Compare with the accepted value and comment.

Worked solution →

[2] Problem 5 — Gradients from a graph. For a pendulum, T^2 is plotted against L. Six points lie close to a straight line with a best-fit gradient of 4.05 {s²,m^(-1)}; the steepest and shallowest acceptable lines have gradients 4.11 and 3.99 {s²,m^(-1)}. (a) Find g and its uncertainty. (b) Find the intercept on the T^2 axis, given the best-fit line passes through (0, 0.02 {s²}). (c) Comment on the intercept.

Worked solution →

[2] Problem 6 — Combining independent uncertainties. Three quantities are measured: x = (10.0 ± 0.5), y = (20.0 ± 0.2) and z = (5.0 ± 0.5). (a) Find f = xyz with its uncertainty. (b) Find h = x + y - z with its uncertainty. (c) Explain why the relative uncertainty in h is smaller than in f even though the absolute uncertainty may be similar.

Worked solution →

[2] Problem 7 — Worst case versus quadrature. Two measurements each have a 2% uncertainty. (a) Find the uncertainty in their product using worst-case addition. (b) Find it using quadrature. (c) Explain when each method is appropriate and why examiners usually accept the quadrature result.

Worked solution →

[3] Problem 8 — The lens experiment. In a lens experiment (1)/(v) is plotted against (1)/(u); the theory gives (1)/(v) = -(1)/(u) + (1)/(f). The best-fit line has gradient -0.98 ± 0.03 and intercept (0.0500 ± 0.0010) {cm^(-1)}. (a) Find f and its uncertainty. (b) Comment on the gradient, which should be exactly -1. (c) Suggest a physical reason for the discrepancy.

Worked solution →

[3] Problem 9 — A resistivity measurement. A wire of length (1.000 ± 0.005) {m} and diameter (0.50 ± 0.01) {mm} has resistance (2.14 ± 0.02) Ω. (a) Find the resistivity. (b) Find its percentage uncertainty. (c) Find how much the uncertainty in the diameter contributes on its own, and comment.

Worked solution →

[3] Problem 10 — Deciding whether results agree. Two students measure the same resistance: (10.2 ± 0.1) Ω and (10.6 ± 0.2) Ω. (a) Find the difference and the combined uncertainty. (b) Decide whether the results are consistent, and express the discrepancy as a multiple of the combined uncertainty. (c) Explain what a third, more precise measurement would settle, and why "they agree within errors" must be justified rather than asserted.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

BIPM: SI brochure and uncertainty guidance

The GUM is the reference behind the uncertainty rules used in 03.02 and 03.03.

Particle Data Group: mathematical tools review

A compact professional summary of means, errors, fits and confidence intervals.